
CONSTELLATION

Concept Paper

Version 2.0 — March 2026

A Universe for Your Knowledge

Multi-vault knowledge management. Built-in databases, tasks, calendars, AI assistance, and templates. Non-destructive. No plugins required.

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Tauri v2 + SvelteKit + Svelte 5 + Rust
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1. What Is Constellation?

Constellation is a desktop knowledge management platform built for people who think in connected notes. It stores everything as standard Markdown files on your local file system — no cloud accounts, no vendor lock-in, no subscription required.

Constellation introduces the Universe — a portable, self-contained workspace that unifies multiple vaults of Markdown files, structured databases, AI assistance, task management, and calendar views into a single coherent experience. Where other tools give you a single notebook, Constellation gives you an interconnected system.

Technical foundation: Tauri v2 (Rust backend) + SvelteKit + Svelte 5. Native performance, small binary size, full offline operation, no Electron overhead.

2. The Problem Constellation Solves

Knowledge management today is fragmented. Users face a common set of problems regardless of which tool they use:

Problem	What Users Do Today	The Cost
Notes scattered across tools	Manual copy-paste between apps	Lost connections, duplicated effort
One notebook/vault at a time	Close one project to open another	Context-switching, no cross-project search or linking
Missing task management	Separate task app (Todoist, Things, etc.)	Tasks disconnected from the notes that created them
No database views	Export to spreadsheets or use separate tools	Data lives outside the knowledge system
Rigid table editing	Edit tables in a spreadsheet, paste back	Workflow interruption, no formulas in notes
Calendar disconnected	Separate calendar app	Daily notes and tasks not visible in one place
Template systems	Manual copy-paste or tool-specific syntax	Inconsistency, wasted setup time

Problem	What Users Do Today	The Cost
AI as an afterthought	Separate AI tool, copy context manually	No integration with your actual notes
Importing from other tools	Manual conversion or format-specific scripts	Friction prevents migration, data stays trapped

Some tools solve a few of these. None solve all of them. Users end up assembling a patchwork of apps, plugins, and workarounds — becoming systems integrators instead of knowledge workers.

Constellation eliminates the integration tax. Every capability listed above is built in, tested together, and ships as a unified experience.

Non-Destructive by Design

Constellation is built on a foundational principle: your files are never modified without your explicit action. It reads your existing Markdown folders exactly as they are — it does not inject metadata, rewrite frontmatter, alter folder structures, or create hidden configuration files inside your vaults. Your Markdown files remain pure, portable, and fully compatible with any text editor or tool that reads standard Markdown.

This means adopting Constellation carries zero risk. Point it at your existing folders of notes, explore every feature, and if you ever decide to use a different tool — nothing has changed. There is no migration, no conversion, and no cleanup required. Constellation is a window into your knowledge, not a lock on it.

3. Core Architecture: The Universe Model

Constellation's defining architectural concept is the Universe — a portable directory that owns all user configuration and workspace state, separate from your notes.

```
MyUniverse/  
universe.json # Identity and metadata  
vaults.json # Registered vault paths  
settings.json # All preferences  
bookmarks.json # Saved bookmarks  
workspaces.json # Tab layouts  
property-types.json # Custom property mappings  
bases/ # Workspace-level databases
```

Why This Matters

- **Portability.** Copy the universe directory to another machine and everything follows — settings, bookmarks, workspaces, database definitions. The vaults themselves are just folders of Markdown files that live wherever you want.
- **Multi-vault by design.** A universe can register any number of vaults. Search, graph view, task scanning, backlinks, and databases all operate across vault boundaries natively.
- **Hierarchy.** Universes can reference child universes, inheriting their vaults. A team lead's universe can include a shared team universe plus a personal universe — with circular reference prevention built in.
- **No lock-in.** The universe is JSON files in a folder. The vaults are Markdown files in folders. Walk away at any time — your notes are standard files that any tool can read.

4. What Constellation Offers

4.1 Capabilities That Set Constellation Apart

Capability	Details
True multi-vault workspace	Open, search, link, and graph across multiple vaults simultaneously in one window.
Universe portability	All configuration travels in a single portable directory. Move machines and your entire workspace follows.
Child universes	Compose workspaces hierarchically — a team vault feeds into your personal universe automatically.
Cross-vault backlinks	See which notes in any vault link to the current note — not limited to a single vault.
Cross-vault graph	One knowledge graph showing connections across all your vaults.
Unified task scanning	Global Tasks view aggregates tasks from every vault with filtering by vault, priority, due date, and text search.
Built-in Bases (databases)	Non-destructive database views with table/card/list modes, filtering, sorting, inline editing — no external tools needed.

Capability	Details
Table formulas	=SUM(), =AVG(), =COUNT(), =MIN(), =MAX() with cell references and ranges, evaluated in-place inside your Markdown tables.
Multi-provider AI	OpenAI, Anthropic, Google Gemini, and Ollama (local) from one interface, with 8 pre-built skills — directly integrated with your notes.
Second screen	A fully independent secondary window that extends your workspace across two screens — edit, browse, view graphs, or manage tasks side by side with no limitations. Not just a reference pane; a complete second workspace.
15 languages at launch	English, Arabic, German, Spanish, French, Hebrew, Hindi, Japanese, Korean, Portuguese, Russian, Turkish, Urdu, Chinese, Farsi — all with full RTL support.
Security layer	Vault encryption at rest, idle lock with PIN, API key storage in OS keyring.
Non-destructive vault access	Never modifies vault files without explicit user action. Zero-risk adoption — try Constellation and switch tools freely with no trace left behind.

4.2 Everything Built In

Features that other tools require plugins, extensions, or external apps to achieve ship built into Constellation:

Feature	How Others Handle It	Constellation (Built-In)
Structured queries	Dataview plugin / external scripts	Native query parser (TABLE, LIST, TASK, CALENDAR queries)
Task management	Separate task apps or plugins	Vault-wide scanning, toggle, due dates, priority, tags
Calendar sidebar	Separate calendar plugins	Month view with note/task dots, daily note creation
Advanced tables	Basic Markdown tables or spreadsheets	Row/column operations, sorting, move, formulas

Feature	How Others Handle It	Constellation (Built-In)
Templates	Manual copy-paste or plugin syntax	Template variables (date, time, title, folder, vault, cursor)
Note importing	Manual conversion scripts	7 formats: Markdown folders, Notion, Bear, Evernote, HTML, CSV, Plain Text
Backlinks panel	Basic or plugin-dependent	Enhanced with cross-vault support and unlinked mentions
Graph view	Single-vault only in most tools	Cross-vault nodes, force controls, grouping
Tag browser	Basic implementations	Tag frequency analysis, vault-wide aggregation

4.3 Import From Anywhere

Constellation's built-in importer supports migration from:

Source	What Gets Imported
Markdown folders	Direct vault registration — no conversion needed
Notion exports	Cleans hex IDs, converts internal links to wikilinks
Bear notes	Converts Bear's format to standard Markdown
Evernote (.enex)	Converts ENML to Markdown, preserves tags and dates as frontmatter
HTML files	Converts to clean Markdown
CSV files	Each row becomes a note with frontmatter properties
Plain text files	Direct import with Markdown extension

Your existing notes from any tool become first-class citizens in Constellation without losing structure or metadata.

4.4 What Constellation Does Not Do (Yet)

Transparency matters. These are capabilities not currently in Constellation:

Feature	Status
Mobile apps (iOS/Android)	Not yet — desktop only (Windows, macOS, Linux)
Cloud sync	Not built-in — use Git, Syncthing, or any file sync solution
Infinite canvas / whiteboard	Not yet
PDF annotation	Not yet
Audio recording	Not yet
Third-party plugin API	Not yet

5. Who Is Constellation For?

5.1 The Multi-Project Professional

Profile: Consultant, researcher, or knowledge worker who maintains separate note collections for different clients, projects, or life domains.

Pain today: Must close one project to open another. Cannot search across collections. Cannot see connections between a client's project notes and research notes in a separate folder.

Constellation answer: Register all vaults in one universe. Search, graph, task scan, and link across all of them simultaneously.

5.2 The Tool-Fatigued Power User

Profile: Power user running multiple apps and extensions who spends significant time managing updates, resolving conflicts, and debugging breakage.

Pain today: Every tool update is a risk. Task management, databases, templates, calendar, and AI are all separate systems maintained by different teams on different schedules.

Constellation answer: All of these are built-in, tested together, and updated as one unit. Zero extension management.

5.3 The Arabic/RTL Knowledge Worker

Profile: User who works primarily in Arabic, Hebrew, Farsi, or Urdu and needs a note-taking system that treats RTL as a first-class concern.

Pain today: RTL support is inconsistent in most tools. Editors assume LTR. Date keys and list keys don't recognize Arabic equivalents. UI elements break in mirrored layouts.

Constellation answer: 15 languages including 4 RTL languages. Arabic property key detection (date, list, checkbox keys recognized in Arabic). Full UI mirroring. RTL-aware tables, forms, editors, and calendar.

5.4 The Team Lead or Organization Builder

Profile: Manager or team lead who wants to share a knowledge base with team members while maintaining personal notes separately.

Pain today: No concept of workspace composition in most tools. Shared note collections require manual setup per person.

Constellation answer: Create a team universe with shared vaults. Each team member adds the team universe as a child of their personal universe. Team vaults appear automatically alongside personal vaults.

5.5 The AI-Augmented Researcher

Profile: Researcher or student who wants AI assistance integrated directly into their note-taking workflow — summarization, Q&A, writing assistance, translation.

Pain today: Must use a separate AI tool, manually copy context, and paste results back. Or install competing AI extensions with inconsistent interfaces and separate API key management.

Constellation answer: One AI settings panel. Four provider options (including local Ollama for privacy). Eight pre-built skills. API keys stored in the OS keyring, not in plaintext config files.

5.6 The Security-Conscious User

Profile: Professional handling sensitive notes (legal, medical, financial, personal) who needs encryption and access control.

Pain today: Most note apps offer no built-in encryption, no idle lock, and store API keys in plaintext configuration files.

Constellation answer: Vault encryption at rest, idle lock with PIN, API key storage in OS keyring.

5.7 The Migrating User

Profile: Someone moving away from Notion, Evernote, Bear, or another tool who wants to own their data locally without losing years of accumulated notes.

Pain today: Migration is painful. Export formats are inconsistent. Internal links break. Metadata gets lost. Many users stay locked in because switching costs are too high.

Constellation answer: Built-in importer handles 7 formats. Notion hex IDs are cleaned, links are converted to wikilinks, Evernote ENML becomes Markdown with frontmatter. One-click migration, zero data loss.

6. Technical Advantages

6.1 Performance

Constellation's Rust backend performs file operations, link scanning, task extraction, and database queries at native speed. Heavy operations — vault-wide task scanning, structured queries, link graph building — execute in the Rust process and return structured results to the frontend. The editor never competes with background processing for resources.

6.2 Binary Size and Resource Usage

Tauri v2 uses the system's native webview rather than bundling Chromium. The result is a significantly smaller binary and lower memory footprint compared to Electron-based alternatives.

6.3 Security Model

Tauri's Rust backend provides a natural security boundary. File system access is controlled through explicit Tauri commands — the frontend cannot access arbitrary files. Path traversal prevention is enforced at the Rust layer (canonicalization checks on all file operations).

6.4 Data Sovereignty

All data lives on the user's file system in standard formats: Markdown files with YAML frontmatter, JSON .base files for databases, JSON files in the universe directory for configuration, and standard image/PDF files in vault folders for attachments. No telemetry. No cloud dependency. No account required.

7. Development Validation Criteria

This section defines how we measure whether Constellation fulfills its purpose. Each criterion maps to a testable capability.

7.1 Core Promise: "Your Notes, Your Way"

Test	Expected Result
Open any folder of Markdown files	All notes visible, frontmatter parsed, links resolved
Edit a note and save	File on disk updates, readable by any Markdown tool
Create a note with frontmatter	Valid YAML frontmatter, standard format
Resolve [[wikilinks]]	Correct resolution across files and folders
Render callouts, highlights, math, mermaid	Rich rendering of extended Markdown syntax

7.2 Multi-Vault Promise: "A Universe of Knowledge"

Test	Expected Result
Register 3+ vaults	All appear in file explorer with distinct colors
Search across vaults	Results from all vaults, labeled by source
Graph across vaults	Nodes from all vaults, cross-vault edges visible
Backlinks across vaults	Note in Vault A shows backlinks from Vault B
Tasks across vaults	Global Tasks view aggregates all vaults

7.3 All-In-One Promise: "Everything Built In"

Test	Expected Result
Structured query in note	TABLE, LIST, TASK queries render results
Task checkbox toggle	Toggle in sidebar updates file on disk
Calendar dot indicators	Days with notes/tasks show visual indicators
Table formula evaluation	=SUM(A1:A5) calculates correctly
Template insertion	Variables replaced with current date, time, title
Import from Notion export	Hex IDs removed, links converted to wikilinks

7.4 User Experience Promise: "Works for Everyone"

Test	Expected Result
Switch to Arabic	Full UI in Arabic, RTL layout, mirrored sidebar
Create note with Arabic properties	Date/list/checkbox keys detected correctly
Open app on new machine with universe copy	All settings, bookmarks, workspaces restored
Lock app, enter PIN	Notes inaccessible until correct PIN entered

8. Competitive Landscape

Constellation occupies a unique position in the knowledge management space: local-first, multi-vault, all-in-one, and multilingual.

Dimension	Constellation	Obsidian	Notion	Logseq	Roam	Bear
Data ownership	Local files	Local files	Cloud-hosted	Local files	Cloud-hosted	iCloud

Dimension	Constellation	Obsidian	Notion	Logseq	Roam	Bear
Offline capability	Full	Full	Limited	Full	None	Full
File format	Standard Markdown	Standard Markdown	Proprietary	Markdown/EDN	Proprietary	Proprietary
Multi-vault	Native (Universe)	One vault per window	N/A (workspaces)	Single graph	Single graph	N/A
Cross-vault search	Yes	No	N/A	No	No	No
Cross-vault graph	Yes	No	N/A	No	No	No
Database views	Built-in (Bases)	Plugin required	Native	Queries (limited)	Queries	No
Task management	Built-in	Plugin required	Basic	Plugin required	Basic	No
AI integration	Built-in(4 providers)	Plugin required	Built-in (1 provider)	Plugin required	Plugin required	No
Table formulas	Built-in	Plugin required	Limited	No	No	No
Import sources	7 formats built-in	Plugin required	Built-in (limited)	Limited	Limited	Limited
RTL / Arabic	15 languages, 4 RTL	Community effort	Limited	Limited	Limited	No
Pricing	Free / Open Source	Freemium	Freemium + subscription	Free	Subscription	Subscription

Dimension	Constellation	Obsidian	Notion	Logseq	Roam	Bear
Architecture	Tauri (Rust + native webview)	Electron	Web app	Electron	Web app	Native (Apple)

Constellation's Position

Constellation does not compete by being "a better version of X." It competes by being a complete knowledge management platform that eliminates the need to assemble a stack of tools. Users coming from any tool — or from no tool at all — can start with Constellation and have everything they need from day one.

For users of existing Markdown-based tools, the transition is seamless: point Constellation at your existing folders and everything works. For users of proprietary tools, the built-in importer handles the conversion.

9. Roadmap

Based on this concept paper, the following development priorities align with Constellation's positioning:

High Priority (Reinforces Core Differentiators)

1. Polish multi-vault experience — cross-vault move/copy, vault-scoped settings
2. Bases performance at scale — handle 10,000+ note databases efficiently
3. AI skill expansion — custom skill builder, context-aware vault Q&A
4. Mobile companion — read-only vault browser for iOS/Android

Medium Priority (Broadens Platform)

5. Canvas / whiteboard — infinite canvas with embedded notes
6. PDF annotation — highlight and annotate PDFs within vaults
7. Publish / static site export — generate websites from vault content
8. Constellation URI protocol — deep linking into specific notes and views

Lower Priority (Future Vision)

-
- 9. Audio recording and transcription
 - 10. Plugin API — allow third-party extensions (carefully scoped)
 - 11. Collaborative editing — real-time multi-user editing via CRDT

10. Conclusion

Constellation exists because knowledge management should not require systems integration. A note-taking platform should ship with the tools its users need — databases, tasks, calendars, templates, importers, AI, and multi-vault support — tested together, updated together, and usable out of the box.

For the knowledge worker who has built a workflow across multiple tools and feels the friction of managing that stack, Constellation offers a unified alternative that works with standard Markdown files, requires zero configuration, and provides capabilities no single existing tool offers — true multi-vault workspaces, cross-vault everything, and portable universe-based configuration.

The files are yours. The format is Markdown. The door is always open.

Constellation is open source under the MIT license.
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Intellectual Property Statement

Constellation is original software developed independently. It does not contain, incorporate, or derive from any third-party application source code. Constellation reads and writes standard Markdown files with YAML frontmatter — open, non-proprietary formats. Wikilink syntax (`[[link]]`) originates from wiki software and is not proprietary to any vendor. File-level interoperability with various Markdown-based tools is achieved through standard file system operations on open formats, not through reverse engineering or use of proprietary APIs.

Open Source Compliance

Constellation is licensed under the MIT License. All third-party dependencies are used in compliance with their respective open source licenses. A full dependency audit is maintained in the project repository.